# Types of Variables

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Before summarizing data, it helps to identify the **type** of each variable, since the appropriate descriptive methods depend on the type. Variables are broadly classified as either **numerical** or **categorical**, and within each class there are important subtypes.

**Definition 0.1** (Numerical variable (quantitative variable))**.** A **numerical** (or **quantitative**) variable is a variable that takes values on a numeric scale, where arithmetic operations such as subtraction (and possibly division) are meaningful.

**Example 0.1** (Numerical variables in the WCGS)**.** Age, systolic blood pressure, total cholesterol, and number of cigarettes smoked per day are numerical variables in the WCGS dataset.

**Definition 0.2** (Interval variable)**.** An **interval variable** is a numerical variable for which differences between values are meaningful, but there is no natural zero point (so ratios of values are not meaningful).

**Example 0.2** (Temperature in degrees Celsius)**.** Temperature in degrees Celsius is an interval variable: a difference of 10 deg C is meaningful, but 30 deg C is not "twice as hot" as 15 deg C because 0 deg C is an arbitrary zero point (the freezing point of water) rather than the complete absence of heat.

**Definition 0.3** (Ratio variable)**.** A **ratio variable** is a numerical variable with a natural zero point, so that both differences and ratios of values are meaningful.

**Example 0.3** (Ratio variables in the WCGS)**.** Age, weight, cholesterol, blood pressure, and number of cigarettes per day are ratio variables: a value of 0 represents the complete absence of the quantity, so ratios of values are meaningful.

**Definition 0.4** (Continuous variable)**.** A **continuous variable** is a numerical variable whose possible values form an interval (or union of intervals) of real numbers. A continuous variable is always numerical.

**Example 0.4** (Continuous variables in the WCGS)**.** Age, systolic blood pressure, total cholesterol, and body mass index (BMI) are continuous variables: their possible values form a continuum of real numbers.

**Definition 0.5** (Discrete variable)**.** A **discrete variable** is a variable whose possible values form a countable set.

**Example 0.5** (Discrete variables in the WCGS)**.** In the WCGS dataset:

- number of cigarettes per day is a discrete numerical variable (count);
- coronary heart disease event status is a discrete categorical variable (binary).

Remark (Subtypes of numerical variables)

*Remark* 0.1 (Subtypes of numerical variables)*.* A numerical variable may be classified in two ways: as an interval variable or a ratio variable, and as a continuous variable or a discrete variable.

Remark (Types of discrete variables)

*Remark* 0.2 (Types of discrete variables)*.* Discrete variables include both numerical types (such as count variables) and categorical types (such as binary, nominal, and ordinal variables). In contrast, continuous variables are always numerical.

**Definition 0.6** (Categorical variable (qualitative variable))**.** A **categorical** (or **qualitative**) variable is a variable that takes values in a finite set of categories, where arithmetic operations such as differences are not meaningful.

**Example 0.6** (Categorical variables in the WCGS)**.** Behavioral pattern (Type A1, A2, B3, B4), smoking status (yes/no), and weight category are categorical variables in the WCGS dataset.

Remark (Kinds of categorical variables)

*Remark* 0.3 (Kinds of categorical variables). Categorical variables may be nominal (unordered) or ordinal (ordered), and are always discrete.

**Definition 0.7** (Nominal variable)**.** A **nominal variable** is a categorical variable whose categories have no natural ordering.

**Example 0.7** (Nominal variables in the WCGS)**.** Behavioral pattern (Type A1, A2, B3, B4), arcus senilis (present/absent), and ABO blood type are nominal variables: there is no natural numerical ordering among the categories.

**Definition 0.8** (Ordinal variable)**.** An **ordinal variable** is a categorical variable whose categories have a natural ordering.

**Example 0.8** (Ordinal variables in the WCGS)**.** Weight category and age group are ordinal variables: their categories have a natural ordering, though the distances between categories are not measured on a numeric scale.

**Definition 0.9** (Binary variable)**.** A **binary variable** takes only two possible values, often coded 0 (absence) and 1 (presence). A binary variable is a special case of a nominal variable.

**Example 0.9** (Binary variables in the WCGS)**.** Coronary heart disease event status (`chd69`, yes/no) and current smoking status (`smoke`, yes/no) are binary variables.

## Coding a categorical variable with numbers

**Exercise 0.1** (Dummy coding)**.** The Western Collaborative Group Study (WCGS) variable `behpat` (behavioral pattern) has four categories: A1, A2, B3, and B4.

(a) How many indicator variables are needed to represent `behpat` if A1 is the reference level?
(b) Give the values of those indicator variables, in the order A2, B3, B4, for a person in category B3, and for a person in category A1.

Solution

*Solution* 0.1.

(a) 3 indicator variables, one for each non-reference category: A2, B3, and B4.

(b) In the order (A2, B3, B4): for B3, the values are (0, 1, 0); for A1, the values are (0, 0, 0).

The reference level is the case where all indicators are 0.

**Definition 0.1** (Dummy variables)**. Dummy variables** are numeric variables that, together, are a numeric representation of a categorical variable.
Source: Dobson and Barnett [1, Sec. 2.4].

**Example 0.1** (Dummy variables for chromosomal sex)**.** Let $S$ be the chromosomal sex of a newborn, a categorical variable with the categories "female" and "male". Let $M = 1$ if $S$ is "male" and $M = 0$ if $S$ is "female", and let $F = 1$ if $S$ is "female" and $F = 0$ if $S$ is "male". Together, $M$ and $F$ are dummy variables for $S$.

Remark (Other ways to code dummy variables)

*Remark* 0.1 (Other ways to code dummy variables). There are other ways to construct dummy variables. One is to use the values $-1$ and 1: for example, $-1$ for "female" and 1 for "male". See Dobson and Barnett [1], section 2.4, for details.

**Definition 0.2** (Indicator variable)**.** An **indicator variable** is a dummy variable whose only values are 0 and 1.
Source: Dobson and Barnett [1, Sec. 2.4].

**Example 0.2** (Indicator variables for chromosomal sex)**.** In Example 0.1, $M$ and $F$ are indicator variables: each takes only the values 0 and 1. A dummy variable that uses the values $-1$ and 1 (Remark 0.1) is not an indicator variable.

**Definition 0.3** (Reference level)**.** The **reference level** of a categorical variable is the category that does not have its own indicator variable. An observation in the reference level has the value 0 for every indicator variable of that categorical variable.

Remark (Source: Reference level)

*Remark* 0.2 (Source: Reference level). Dobson and Barnett [1, Sec. 2.4], who call this the corner point parameterization.

**Example 0.3** (The reference level for chromosomal sex)**.** Take chromosomal sex $S$ with the indicator variable $M$ for "male" (Example 0.1). The category "female" has no indicator variable, so "female" is the reference level, and a female newborn has $M = 0$. In Exercise 0.1, category A1 of `behpat` is the reference level.

Remark (Any category can be the reference level)

*Remark* 0.3 (Any category can be the reference level). The choice of reference level changes how we code the categories. It does not change what the data say. For example, in Example 0.3, we could instead use "male" as the reference level and the indicator variable $F$ for "female". Both choices describe the same two groups.

## Taxonomy of variable types

Figure 1 illustrates the relationships among these variable types.

Note

The continuous/discrete distinction cuts across the numerical/categorical distinction. Continuous variables are always numerical. Discrete variables include both numerical types (such as count variables) and categorical types (such as binary, nominal, and ordinal variables).

## Variables in the WCGS dataset

Table 1 shows selected variables from the Western Collaborative Group Study (WCGS) dataset and their types.

```
tibble::tribble(
  ~Variable, ~Description, ~Type, ~Scale,
  "`age`", "Age (years)", "Continuous", "Ratio",
  "`chol`", "Total cholesterol", "Continuous", "Ratio",
  "`sbp`", "Systolic blood pressure", "Continuous", "Ratio",
```

```r
nodes <- tibble::tribble(
  ~id,   ~x,    ~y,   ~label,
  "V",    5,    4.5,  "Variables",
  "N",    2.5,  3,    "Numerical\n(quantitative)",
  "C",    7.5,  3,    "Categorical\n(qualitative)",
  "I",    1,    1.5,  "Interval\n(no true zero)\ne.g. temp. in deg C",
  "R",    4,    1.5,  "Ratio\n(true zero)\ne.g. age, weight",
  "CT",   3,    0,    "Continuous\ne.g. age, BMI",
  "CNT",  5,    0,    "Count\n(discrete)\ne.g. cigs/day",
  "NOM",  6.5,  1.5,  "Nominal\n(unordered)\ne.g. blood type",
  "ORD",  8.5,  1.5,  "Ordinal\n(ordered)\ne.g. wt. category",
  "BIN",  6.5,  0,    "Binary\n(2 categories)\ne.g. CHD event"
)
edges <- tibble::tribble(
  ~from, ~to,
  "V",   "N",
  "V",   "C",
  "N",   "I",
  "N",   "R",
  "R",   "CT",
  "R",   "CNT",
  "C",   "NOM",
  "C",   "ORD",
  "NOM", "BIN"
) |>
  dplyr::left_join(
    dplyr::select(nodes, id, x, y),
    by = c("from" = "id")
  ) |>
  dplyr::rename(x_from = x, y_from = y) |>
  dplyr::left_join(
    dplyr::select(nodes, id, x, y),
    by = c("to" = "id")
  ) |>
  dplyr::rename(x_to = x, y_to = y)
fill_colors <- c(
  "V" = "#f0f0f0",
  "N" = "#d0e8ff", "C" = "#ffe8d0",
  "I" = "#e8f4ff", "R" = "#e8f4ff",
  "CT" = "#c8e8ff", "CNT" = "#c8e8ff",
  "NOM" = "#ffe0c0", "ORD" = "#ffe0c0",
  "BIN" = "#ffd0a0"
)
ggplot2::ggplot() +
  ggplot2::aes() +
  ggplot2::geom_segment(
    data = edges,
    ggplot2::aes(
      x = x_from, y = y_from - 0.45,
      xend = x_to, yend = y_to + 0.45
    ),
    color = "grey50"
  ) +
```

```
  "`bmi`", "Body mass index (kg/m^2)", "Continuous", "Ratio",
  "`weight`", "Weight (lbs)", "Continuous", "Ratio",
  "`ncigs`", "Cigarettes per day", "Count (discrete)", "Ratio",
  "`chd69`", "CHD event by 1969", "Binary (nominal)", "Nominal",
  "`smoke`", "Current smoking", "Binary (nominal)", "Nominal",
  "`arcus`", "Arcus senilis", "Binary (nominal)", "Nominal",
  "`dibpat`", "Behavioral pattern (A/B)", "Binary (nominal)", "Nominal",
  "`behpat`", "Behavioral pattern (A1/A2/B3/B4)", "Nominal", "Nominal",
  "`wghtcat`", "Weight category", "Ordinal", "Ordinal",
  "`agec`", "Age group", "Ordinal", "Ordinal"
) |>
  knitr::kable()
```

Table 1: Selected WCGS variables and their types

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| Variable | Description | Type | Scale |
|---|---|---|---|
| `age` | Age (years) | Continuous | Ratio |
| `chol` | Total cholesterol | Continuous | Ratio |
| `sbp` | Systolic blood pressure | Continuous | Ratio |
| `bmi` | Body mass index (kg/m^2) | Continuous | Ratio |
| `weight` | Weight (lbs) | Continuous | Ratio |
| `ncigs` | Cigarettes per day | Count (discrete) | Ratio |
| `chd69` | CHD event by 1969 | Binary (nominal) | Nominal |
| `smoke` | Current smoking | Binary (nominal) | Nominal |
| `arcus` | Arcus senilis | Binary (nominal) | Nominal |
| `dibpat` | Behavioral pattern (A/B) | Binary (nominal) | Nominal |
| `behpat` | Behavioral pattern (A1/A2/B3/B4) | Nominal | Nominal |
| `wghtcat` | Weight category | Ordinal | Ordinal |
| `agec` | Age group | Ordinal | Ordinal |

# Random variables

## Binary variables

**Definition 0.1** (Binary random variable)**.** A **binary variable** is a random variable which has only two possible values in its range.

**Exercise 0.1** (Examples of binary variables)**.** What are some examples of binary variables in the health sciences?

Solution

*Solution* 0.1. Examples of binary outcomes include:

- exposure (exposed vs unexposed)
- disease status (diseased vs healthy)
- recovery (recovered vs unrecovered)
- relapse (relapse vs remission)
- return to hospital (returned vs not returned)
- vital status (dead vs alive)

## Count variables

**Definition 0.2** (Count variable)**.** A **count variable** is a random variable whose possible values are some subset of the non-negative integers; specifically, a random variable $X$ such that:

$$\mathcal{R}(X) \subseteq \mathbb{N}$$

**Exercise 0.2** (Examples of count variables)**.** What are some examples of count variables?

Solution

*Solution* 0.2. Examples of count variables include:

- number of fish in a pond
- number of cyclones per season
- seconds of tooth-brushing per session (when rounded)

- infections per person-year
- visits to an emergency room per person-month
- car accidents per 1,000 miles driven

### Probability distributions for count outcomes

Standard probability distributions for count outcomes include:

- Poisson distribution
- Negative binomial distribution

## References

[1] A. J. Dobson and A. G. Barnett, *An introduction to generalized linear models*, 4th ed. CRC press, 2018. doi: 10.1201/9781315182780.